

Executive Summary: Customer Churn Analysis

1. The Core Business Problem

Our dataset shows a macro churn rate of 10.60% (meaning we lost 53 out of 500 total customers). While losing 10% of customers might sound small, looking closely at the data reveals that the customers who left were our highest-paying subscribers.

Losing these accounts means the company is losing:

- \$6,878.00 every single month in recurring revenue.
- \$82,536.00 in total value over a year.

Keeping these customers is much cheaper than spending marketing money trying to find new ones to replace them.

2. What the Data Tells Us (Exploratory Findings)

By looking at the differences between customers who stayed and customers who left, we found two massive red flags:

- The Price Barrier: Customers who quit were paying much higher monthly rates on average than customers who stayed. This points to "price fatigue"—the service became too expensive for them.
- The Contract Trap: A massive majority of the customers who left were on flexible Month-to-month contracts. Customers locked into One-year or Two-year contracts rarely left.

3. Proving It with Math (Statistical Validation)

We ran two main tests to ensure these patterns were real and not just a random coincidence:

- The Pricing Test (T-Test): We compared the exact monthly bills. Retained customers averaged \$111.72 per month, while churned customers averaged \$129.77. The test confirmed with 95%+ statistical confidence that this price gap is exactly what is driving customers away.
- The Risk Test (Regression): We ran a predictive model to see how pricing, contract types, and loyalty (tenure) interact at the same time. The math proved that:
 1. Every extra month a customer stays with us naturally lowers their risk of leaving (building loyalty).
 2. Signing a customer onto a One-year or Two-year contract is the single strongest shield we have against attrition.

4. Actionable Next Steps

Based on these findings, we recommend three simple strategies to save revenue:

1. Target the 6-Month Mark: The data shows customers are most likely to drop off in their first six months. We should set up automatic email check-ins or offer a support call at month 5 to help them past this hurdle.
2. Push for 1-Year Contracts: Run a promotion offering a small discount (like 10% off) if a Month-to-month customer switches to a stable 1-year contract. The locked-in security far outweighs the minor discount.
3. Protect Premium Users: For accounts facing very high monthly bills, bundle in free add-on services or loyalty perks. This increases their perceived value so they don't feel overcharged.